

SmartTrim™ MEMS Microphone with I2S/TDM for Microphone Array

DESCRIPTION

The ZTS6672S is a high quality, high performance, low power digital output bottom-ported omni-directional MEMS microphone. ZTS6672S consists of a MEMS microphone element and a preamplifier. ZTS6672S has a high SNR and flat wideband frequency response, resulting in natural sound with high intelligibility. Due to built-in filter, ZTS6672S shows high immunity to EMI. The ZTS6672S is available in a thin 3.50mm × 2.65mm × 0.98mm surface-mount package. It is reflow solder compatible with no sensitivity degradation. The ZTS6672S is Halogen and Lead free.

APPLICATIONS

- Small portable devices, wearables
- Set-top boxes, TV, gaming, remote controllers
- Smart home devices, Internet of Things, Connected equipment

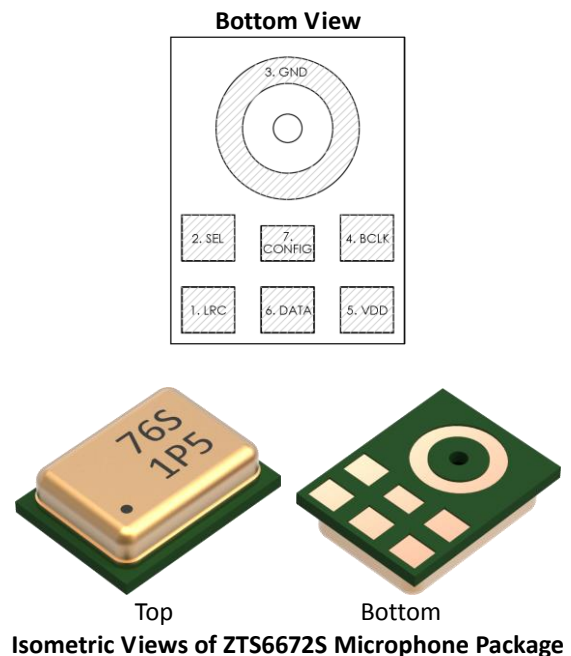
ORDERING INFORMATION

PART	RoHS	Ship, Quantity
ZTS6672S	Yes	Tape and Reel, 5.2K

FEATURES

- Audio data Slave I2S or time division multiplexed (TDM) output interface
- Configurable TDM slots, support up to 8 format
- Supply operation: 1.60 V to 3.60V
- 64×/128×/192×/256×/384×/512× output sample rate BCLK
- SNR of 64 dB(A)
- Sensitivity of -26 dB FS
- Multi-Chip Module (MCM) Package

Pins Configuration and Description



Typical Applications

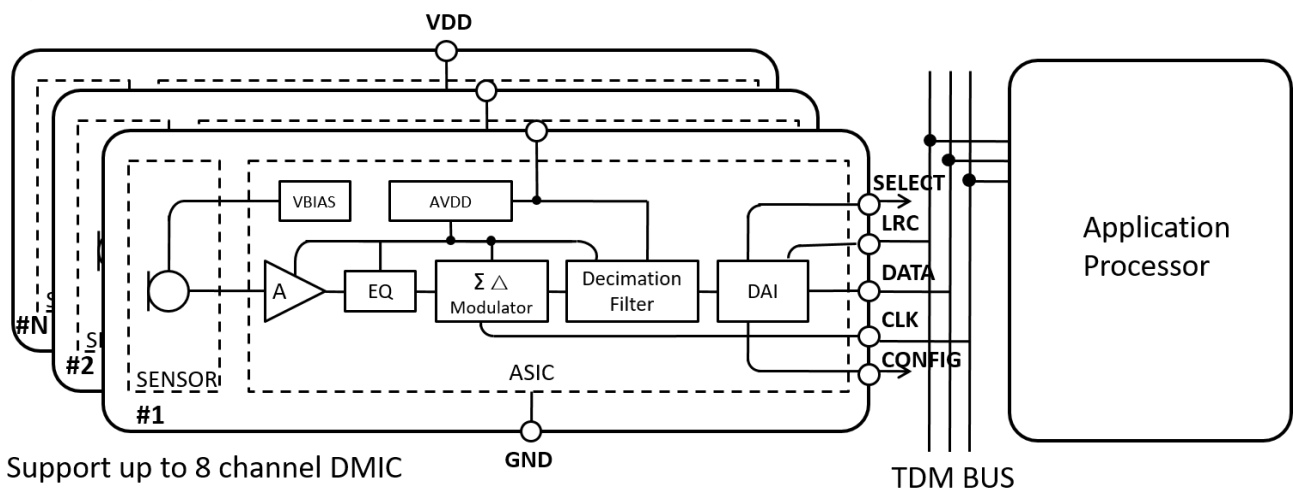


Figure.1 Application Diagram

Absolute Maximum Ratings

CLOCK to Ground -0.3V to +5.0V
 SELECT, V_{DD}, DATA to Ground -0.3V to +5.0V
 Input Current ±5mA
 Data Output Short Circuit Indefinite to Ground or V_{DD}
 Operating Temperature Range -40°C to +125°C
 Storage Temperature Range -40°C to +125°C

CAUTION: Stresses above those listed in “Absolute Maximum Ratings” may cause permanent damage to the device. This is a stress only rating and operation of the device at these or any other conditions above those indicated in the operational sections of this specification is not implied.

Electro-Static Discharge Sensitivity

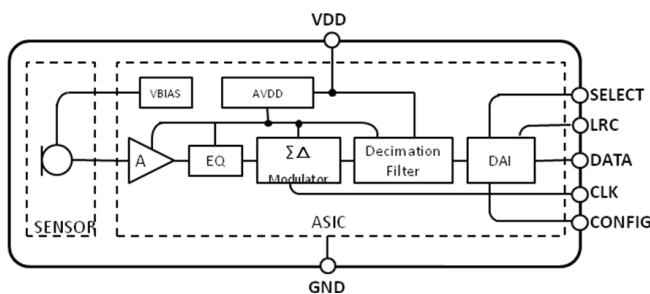


This integrated circuit can be damaged by ESD. It is recommended that all integrated circuits be handled with proper precautions. Failure to observe proper handling and installation procedures can cause damage. ESD damage can range from subtle performance degradation to complete device failure.

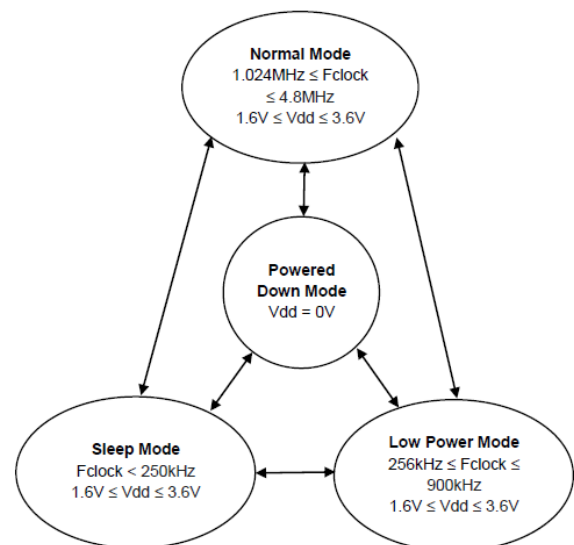
Pins Description

Pin	Symbol	Description
1	LRC	Frame Sync
2	SELECT	Select.
3	GND	Ground
4	BCLK	Bit Clock
5	VDD	Power Supply (VDD).
6	DATA	Output.
7	CONFIG	Interface Configuration

Microphone Block Diagram



Microphone State Diagram



Specifications

(TA = +23°C, VDD = +1.8V, fclock = 3.072MHz, R.H. = 60%~70%, no load, VDD Decoupling cap=1.0uF unless otherwise noted.)

PARAMETER	Symbol	TEST CONDITIONS	MIN	TYP	MAX	UNIT
Supply Voltage (Note 1)	VDD		1.60		3.60	V
Current Consumption (Note 1,5,6)	IDD			800	1000	μA
Sleep Current (Note 6)	ISLEEP	fclock < 1KHz			30	μA
Directivity			Omni-directional			
Sensitivity (Note 1)	S	94dB SPL @ 1KHz	-27	-26	-25	dB FS
Signal to Noise Ratio	SNR	94dB SPL @ 1KHz, A-weighted		64		dB(A)
Total Harmonic Distortion	THD	94dB SPL @ 1KHz			0.12	%
		116dB SPL @ 1KHz			1	%
Acoustic Overload Point	AOP	10% THD @ 1 kHz, S = Typ.		120		dB SPL
Power Supply Rejection Ratio	PSRR	1KHz, 200mV Vp-p, sine wave on VDD		60		dBV/FS
Power Supply Rejection	PSR+N	217Hz, 100mV Vp-p, square wave on VDD		-86		dB FS(A)
Short Circuit Output Current	ISC	Grounded output pin	2		10	mA
Output Load	CLOAD				120	pF
Data Format			24bits			
Clock Frequency	fclock		1.024		4.8	MHz
Low power Clock Frequency	fsleep				900	KHz
Clock Duty Cycle			40		60	%
Clock Rise/Fall Time	tedge				13	ns
Logic Input Low	VIL		-0.3		0.35×VDD	V
Logic Input High	VIH		0.65×VDD		VDD+0.3	V
Logic Output Low	VOL	Iout = 2mA	0		0.3×VDD	V
Logic Output High	VOH	Iout = 2mA	0.7×VDD		VDD	V
Low→High Threshold	VL-H		0.55×VDD		0.65×VDD	V
High→Low Threshold	VH-L		0.35×VDD		0.45×VDD	V
Hysteresis Width	VHYST		0.10×VDD		0.29×VDD	V
Functional Temperature		Functional with lower performance	-40		100	°C
Operating Temperature		Specifications guaranteed	0		45	°C
Power-up Time (Note 4)		VDD ≥ V(min)		13	50	ms
SELECT (high)			VDD -0.2		VDD	V
SELECT (low)		GND			GND+0.2	V
Select Input	CSELECT				2	pF
Clock Input	CCLK				2	pF
Clock Duty Cycle			40	50	60	%
TIE Clock Jitter	TIE	Time Interval Jitter on CLK line			2	ns RMS

(TA = +23°C, VDD = +1.8V, fclock = 768KHz, R.H. = 60%~70%, no load, VDD Decoupling cap=1.0uF unless otherwise noted.)

PARAMETER	Symbol	TEST CONDITIONS	MIN	TYP	MAX	UNIT
Supply Voltage (Note 1)	V _{DD}		1.6		3.6	V
Clock Frequency	Fclock		256		900	KHz
Current Consumption (Note 1,5,6)	IDD			370	450	μA
Sleep Current (Note 6)	ISLEEP	fclock < 1KHz			30	μA
Directivity			Omni-directional			
Sensitivity (Note 1)	S	94dB SPL @ 1KHz	-27	-26	-25	dB FS
Signal to Noise Ratio	SNR	94dB SPL @ 1KHz, A-weighted		64		dB(A)
Total Harmonic Distortion	THD	94 dB SPL @ 1KHz			0.12	%
Acoustic Overload Point	AOP	10% THD @ 1 kHz, S = Typ.		120		dB SPL
Power Supply Rejection Ratio	PSRR	1KHz, 200mV Vp-p, sine wave on V _{DD}		45		dBV/FS
Power Supply Rejection	PSR+N	217Hz, 100mV Vp-p, square wave on V _{DD}		-76		dB FS(A)
Short Circuit Output Current	ISC	Grounded output pin	2		10	mA
Output Load	CLOAD				120	pF
Data Format			24bits			
Clock Duty Cycle			40		60	%
Sleep Clock Frequency	fsleep				1	KHz
Clock Rise/Fall Time	tedge				13	ns
Logic Input Low	VIL		-0.3		0.35×VDD	V
Logic Input High	VIH		0.65×VDD		VDD+0.3	V
Logic Output Low	VOL		0		0.3×VDD	V
Logic Output High	VOH		0.7×VDD		VDD	V
Low→High Threshold	VL-H		0.55×VDD		0.65×VDD	V
High→Low Threshold	VH-L		0.35×VDD		0.45×VDD	V
Hysteresis Width	VHYST		0.10×VDD		0.29×VDD	V
Functional Temperature		Functional with lower performance	-40		100	°C
Operating Temperature		Specifications guaranteed	0		45	°C
Power-up Time ^{4,5}		VDD ≥ V(min)			50	ms
SELECT (high)			VDD -0.2		VDD	V
SELECT (low)		GND			GND+0.2	V
Select Input	CSELECT				2	pF
Clock Input	CCLK				2	pF
Clock Duty Cycle			40	50	60	%
TIE Clock Jitter	TIE	Time Interval Jitter on CLK line			2	ns RMS

Note 1: 100% tested.

Note 2: Valid microphone states are: Power Down Mode (mic off), Sleep Mode (low current, no output, fast start-up), and Active Mode (normal operation).

Note 3: Time from fclock < 250KHz to sleep current specification is met when transitioning from low power mode to Sleep Mode.

Note 4: Time from fclock ≥ 1MHz to all applicable specifications are met when transitioning from low power mode to normal Mode.

Note 5: ΔIDD = 0.5 × VDD × CLOAD × fclock

Note 6: Specified max values are measured at VDD = +3.6V.

Timing Diagram

In I2S mode, the MSB of the left channel is valid on the second rising edge of the bit clock after the falling edge of the LR clock. Similarly the MSB of the right channel is valid on the second rising edge of the bit clock after the rising edge of the LRclock.

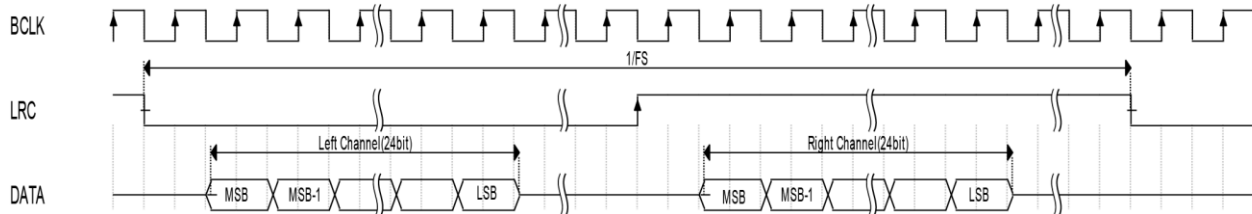


Figure 1

TDM mode

In TDM mode, the falling edge of the LR clock starts the data transfer with the slot 0 data first and immediately followed by the slot1 data. Each data bit is valid on the rising edge of the bit clock.

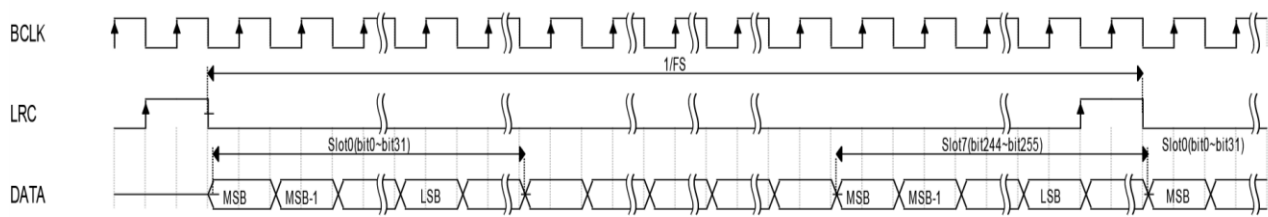


Figure 2

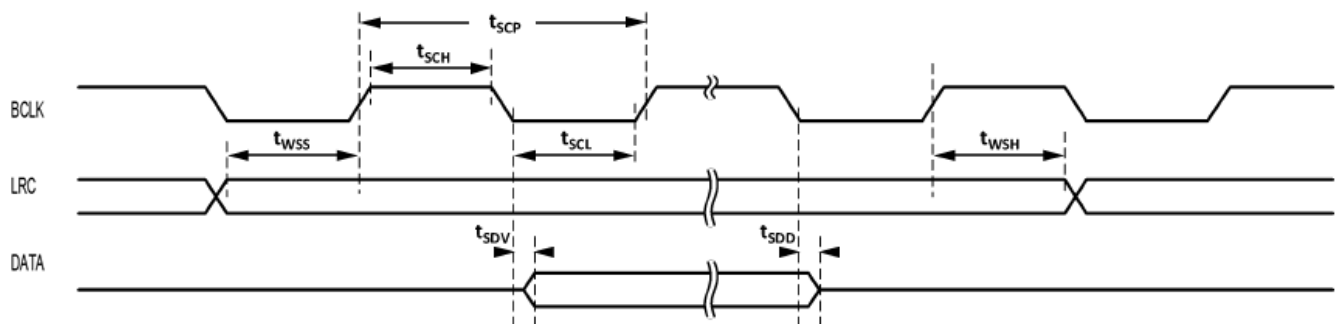
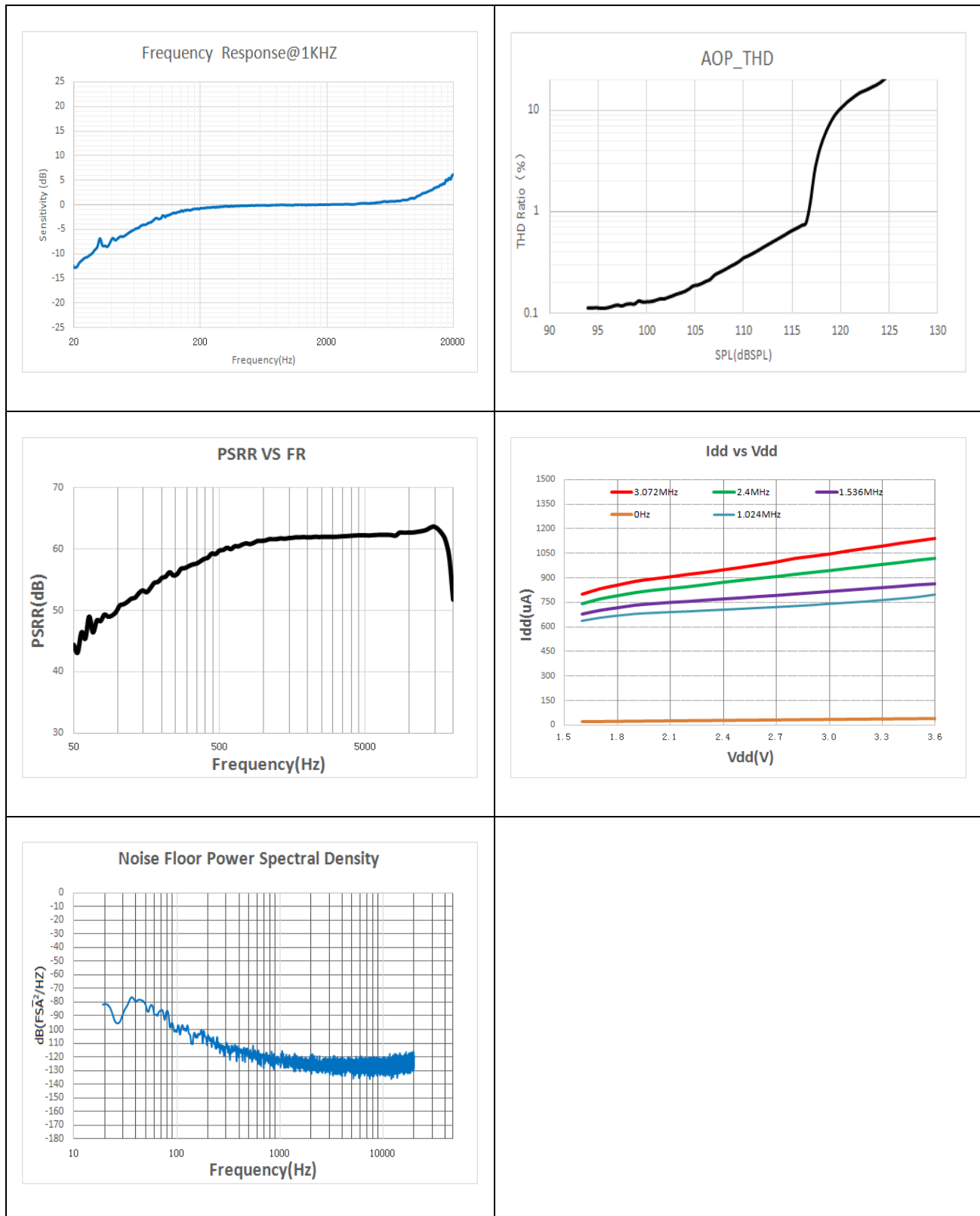


Figure 3

Parameter	Description	Min.	Norm.	Max.	Unit
tSCH	BCLK High	75	—	—	ns
tSCL	BCLK Low	75	—	—	ns
tSCP	LRC Period	208	325	2500	ns
fSCK	BCLK Frequency	0.256	3.072	4.8	MHz
tWSS	LRC Setup	0	—	—	ns
tWSH	LRC Hold	20	—	—	ns
tSDV	From BCLK falling to valid data	—	—	75	ns
tSDD	From BCLK falling to output tristated	—	—	76	ns

Typical Performance Characteristics



Application Notes

SERIAL AUDIO OUTPUT INTERFACE

The ZTS6672S supports I2S and TDM serial output formats. Format selection and TDM slot placement is set with the CONFIG pin. The SDATA pin is in tristate mode, except when the port is driving serial data based on the CONFIG pin configuration.

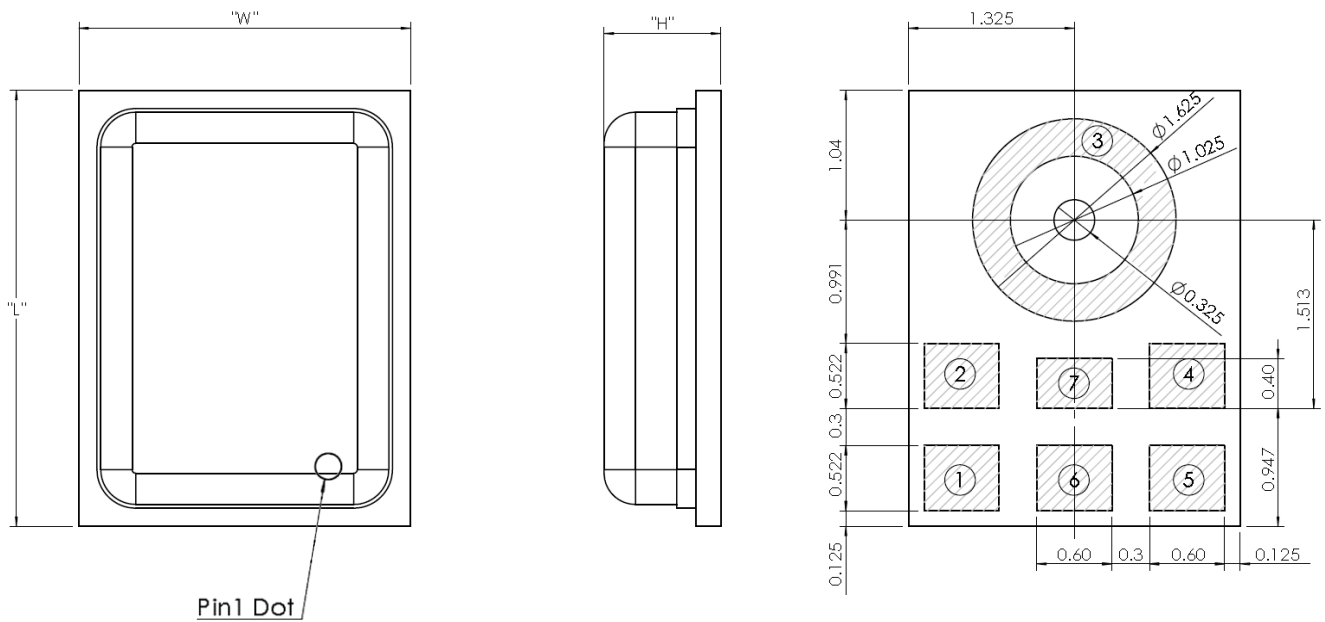
CONFIG	SELECT	DEVICE SETTING
Float	Pull High	I2S Format Right Channel
	Pull Low	I2S Format Left Channel
Pull Low	Pull Low	TDM Slot 1
	Pull High	TDM Slot 2
Pull High	Pull Low	TDM Slot 3
	Pull High	TDM Slot 4
Pull High with 47K ohm	Pull Low	TDM Slot 5
	Pull High	TDM Slot 6
Pull Low with 47Kohm	Pull Low	TDM Slot 7
	Pull High	TDM Slot 8

Reliability Tests

The microphone sensitivity after stress must deviate by no more than $\pm 3\text{dB}$ from the initial value.

1. Heat Test, Operational	Temperature: $125\pm 3^{\circ}\text{C}$ Duration: 1000 hours Voltage: Applied
2. Cold Test, Operational	Temperature: $-40\pm 3^{\circ}\text{C}$ Duration: 1000 hours Voltage: Applied
3. Heat Test, Non-Operational	Temperature: $125\pm 3^{\circ}\text{C}$ Duration: 1000 hours Voltage: Not Applied
4. Cold Test, Non-Operational	Temperature: $-40\pm 3^{\circ}\text{C}$ Duration: 1000 hours Voltage: Not Applied
5. Thermal Shock Test, Non-Operational	Temperature: $-40\pm 3^{\circ}\text{C}$ and $125\pm 3^{\circ}\text{C}$ Duration: 30 minutes each, during 5 minutes ramp, 256 cycles Voltage: Not applied
6. Temperature humidity storage	Temperature: $85\pm 3^{\circ}\text{C}$ Humidity: $85\pm 3\%\text{RH}$ Duration: 1000 hours
	Temperature: $65\pm 3^{\circ}\text{C}$ Humidity: $95\pm 3\%\text{RH}$ Duration: 168 hours
7. Free Fall Test 1.5m	Placed inside test fixture and dropped on concrete from height 1.5m. 4 times by each surface and corner
8. Vibration	4 cycles of 20 to 2000 Hz sinusoidal sweep with 20G peak acceleration lasting 12 minutes in X, Y, and Z directions
9. Mechanical Shock	5 pulses of 10000g in each of the $\pm X$, $\pm Y$, and $\pm Z$ directions
10. Electrostatic Discharge Test	Capacitance: 150pF Resistance: 330 Ω Duration: 10 times Air Discharge: Level 4(+/-15kV) Direct contact discharge: Level 4 (+/-8kV)
11. Human Body Mode	± 5000 Volt
12. Charged-Device Model	± 500 Volt
13. Reflow	5 reflow cycles with peak temperature of 260°C
14. Solderability	$245\pm 5^{\circ}\text{C}$, 5sec, 95% Tin on pad surface
15. Tumble test	300 tumbles from a height of 1m onto a steel base.
16. HAST	Temperature: $130\pm 3^{\circ}\text{C}$ Humidity: $85\pm 3\%\text{RH}$ Duration: 96 hours Voltage: Applied
17. Air Blow	0.45MPa, distance 3cm, time 10s

MECHANICAL SPECIFICATIONS

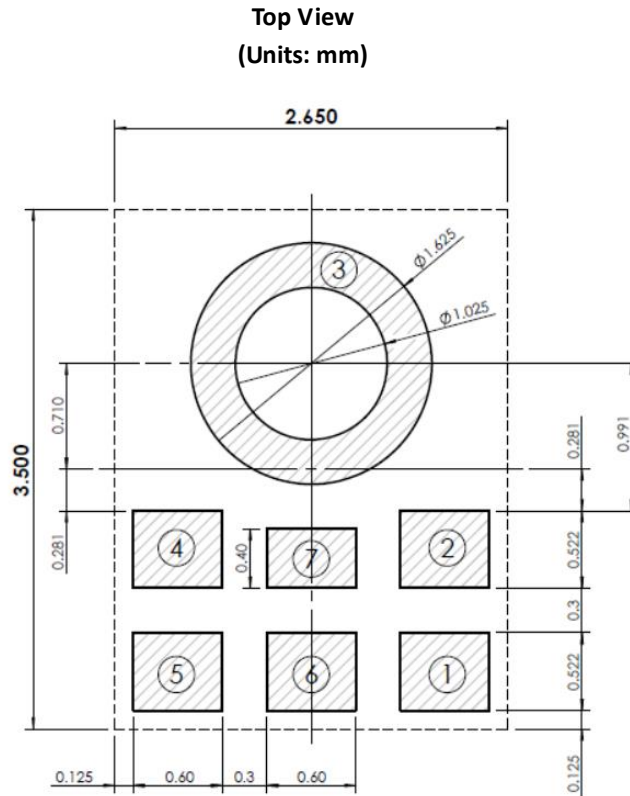


ITEM	DIMENSION	TOLERANCE	UNITS
Length (L)	3.50	±0.10	mm
Width (W)	2.65	±0.10	mm
Height (H)	0.98	±0.10	mm
Acoustic Port (AP)	Ø0.325	±0.05	mm

Pin	Symbol	Description
1	LRC	Frame Sync
2	SELECT	Select.
3	GND	Ground
4	BCLK	Bit Clock
5	VDD	Power Supply (VDD).
6	DATA	Output.
7	CONFIG	Interface Configuration

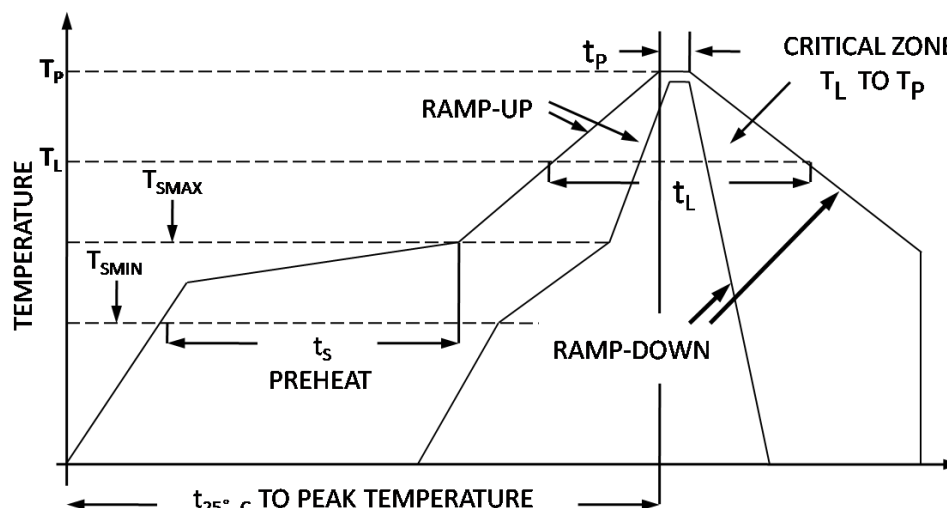
RECOMMENDED CUSTOMER LAND PATTERN

The recommended PCB land pattern for the ZTS6672S should have a 1:1 ratio to the solder pads on the microphone package. Care should be taken to avoid applying solder paste to the sound hole in PCB. The dimensions of suggested solder paste pattern refer to the land pattern which should be shrunk by 0.025 per side.



SOLDER REFLOW PROFILE

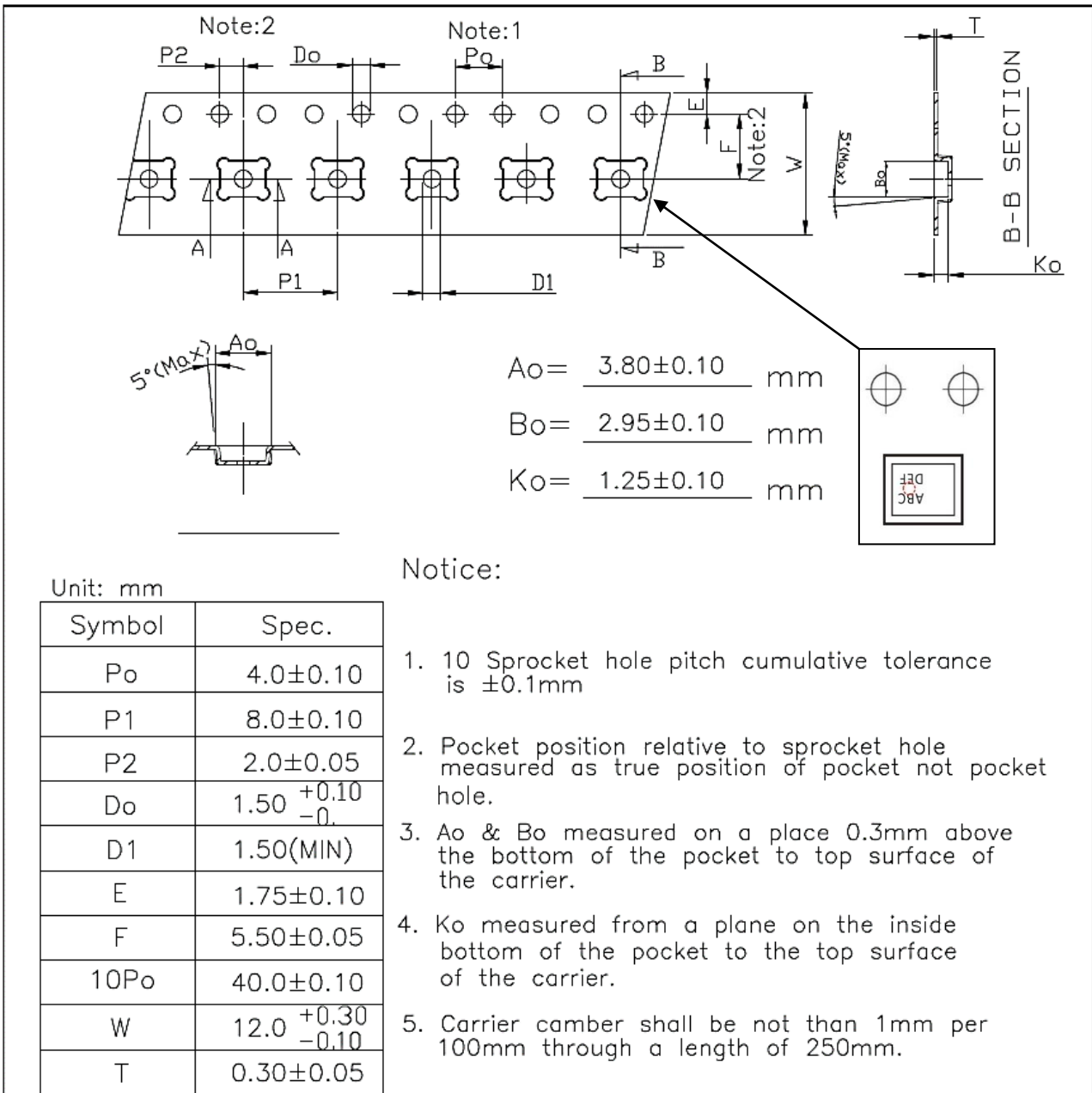
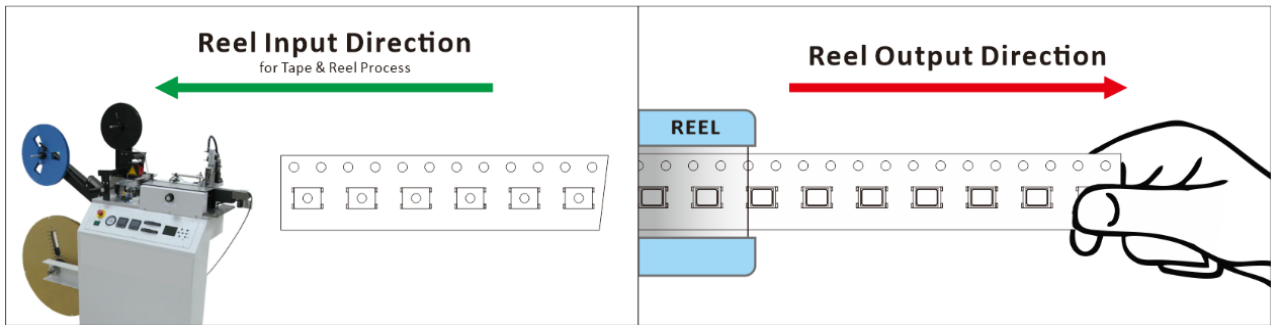
The reflow profile specified in this section describes expected maximum heat exposure of components during the reflow process of NMP product PWBs. Temperature is measured on top of component. All components have to tolerate at least this profile five times (5x) without affecting electrical performance, mechanical performance or reliability.

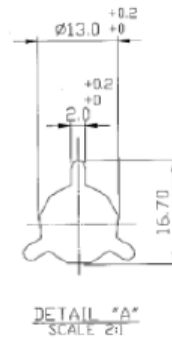
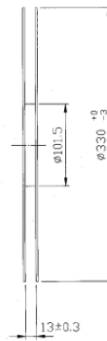
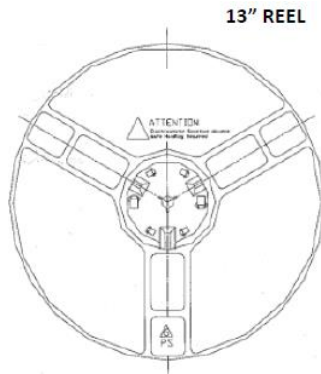


Pb-free reflow profile requirements for soldering heat resistance:

Parameter		Reference	Pb-Free
Average Ramp Rate		T_L to T_P	3°C/sec max
Preheat	Minimum Temperature	T_{SMIN}	150°C
	Maximum Temperature	T_{SMAX}	200°C
	Time	T_{SMIN} to T_{SMAX}	60sec to 180sec
Time Maintained Above	Temperature	T_L	217°C
	Time	t_L	60sec to 150sec
Peak Temperature		T_P	260°C
Time Within +5°C of Actual Peak Temperature		t_p	20 sec to 40 sec
Ramp-Down Rate		T_{peak}	6°C/sec max
Time +25°C (t_{250C}) to Peak Temperature			8 min max

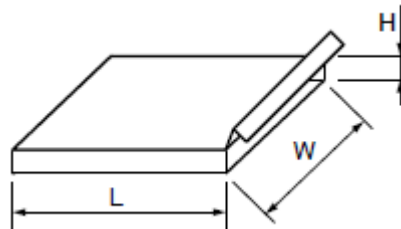
PACKAGING





Part NO.	Reel Diameter	Quantity Per Reel	Quantity Per Inner Box	Quantity Per Outer Box
ZTS6672S	13"	5200	5200	46800

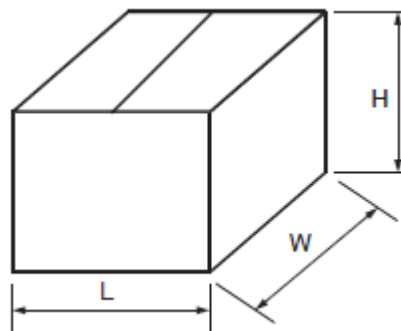
Dimensions for Inner Box



Unit : mm

L	W	H
335	339	44

Dimensions for Outer Box



Unit : mm

L	W	H
445	360	372